

Andy Zheng

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SKILLS

- SQL: PostgreSQL, MySQL, SQL Server
- TABLEAU: Data Visualization, Dashboards
- EXCEL: Pivot Tables, Conditional Formatting, Xlookup, Dashboards
- PYTHON: Pandas, NumPy, Matplotlib, Seaborn
- ECONOMICS: Statistics, Time-Series, Hypothesis Testing, Probability, Regression

PROJECTS & EXPERIENCE

BANK TRANSACTIONS ANALYSIS

February 2025

Project

- Cleaned and transformed >1M rows of banking transaction data using SQL & PowerQuery to uncover actionable insights and ensure data consistency and reliability.
- Built Interactive Tableau dashboards visualizing \$570M in transaction volume across 13.3M transactions with core banking KPIs (e.g. Transaction Volume, DTI, SIR), financial behavior and fraud & error analysis.
- Conducted detailed analysis on transaction patterns and fraud cases, uncovering \$1.47M in fraudulent activity and identifying high risk categories.
- Provided 15+ strategic recommendations to enhance banking operations, including targeted solutions for the 123,552 "Insufficient Balance" errors and key demographic areas.

ELECTRONIC SALES PERFORMANCE

January 2025

Project

- Used SQL, Excel and Tableau to analyze a year's worth of electronic sales data revealing seasonal revenue peaks and identifying top performing products. Evaluated customer loyalty program and discovered opportunities for growth.
- Developed RFM (Recency, Frequency, Monetary) analysis to segment customers, uncovering a strong correlation between scores and spending, with mid-tier groups emerging as top revenue contributors. Developed targeted retention strategies to boost loyalty program enrollment and maximize customer lifetime value.
- Created performance dashboards visualizing 100K+ transaction records, including sales trends, customer segmentation, product and add-on analysis.

TRAVEL & AIRLINE OPERATIONS REPORT

December 2024

Personal Project

- Conducted analysis of airline operations, examining 33,121 flights to optimize scheduling and enhance operational efficiency. Provided insights into flight delays, booking patterns, and aircraft performance.
- Designed an interactive dashboard to visualize revenue trends, passenger preferences, and flight occupancy rates, enabling data-driven strategies.
- Identified passenger booking trends and preferences, highlighting business-class demand on long-haul routes and peak travel times which aligned with strategies for increasing revenue and optimizing customer experiences.

ECOMMERCE CONSUMER BEHAVIOR

October 2024

Project

- Analyzed consumer behavior across 275K+ transactions on an E-commerce platform, examining purchasing patterns and demographics to design targeted campaigns for increasing customer engagement.
- Utilized Excel to design a dynamic dashboard and performed RFM analysis to segment customers and uncover high-value groups, enabling data-driven marketing strategies that improved revenue targeting.

FREELANCE

June 2023 - Present

Data Entry & Ad-Hoc Projects

- Performed data entry tasks to organize and maintain spreadsheets and records.
- Handled ad-hoc requests, including data organization and research & collection, tailored to individual client needs.

EDUCATION

STONYBROOK UNIVERSITY

Long Island, NY

Bachelor of Economics

Relevant Coursework: Data Science in R, Econometrics, Business Analytics, Risk Management, Corporate Finance